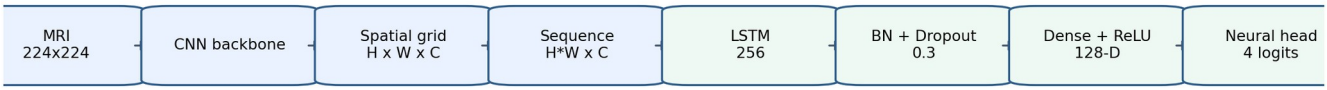
DL TUMOR CLASSIFICATION

Comprehensive Analysis, Phase 2A/2B Findings, and Project Transfer Record

Brain Tumor MRI Classification | Status snapshot: 2 October 2026



Deep features -> StandardScaler (fit train only) -> PCA(64, fit train only) -> Linear SVM

5	40	35/35	1,061
backbones	fixed 30-epoch runs	Phase 2B variants complete	material-clean test images

Scientific interpretation gate

The original train/validation/test arrangement contains material cross-split image duplication. Original scores are historical contaminated-split results. The material_clean test evaluation is a post-hoc leakage-aware check, but patient-level independence remains unknown and duplicate-group-aware retraining has not yet been performed.

Purpose: a self-contained record for future ChatGPT/Claude work, report writing, slide generation, and methodological continuity.

How to use this document

This report separates verified numerical evidence from project-level status decisions. Sections 8-10 are recalculated directly from the uploaded Phase 2A/2B JSON result files. Earlier Phase 1, leakage, material-clean, GUI, and explainability findings are preserved from the established project record and source artifacts.

Evidence tier	What it contains	Use
Tier A - directly re-verified now	5 Phase 2A baseline JSONs + 35 Phase 2B JSONs; 30-epoch histories and configurations	Primary source for Phase 2A/2B tables, deltas, sensitivity ranges and stability analysis
Tier B - project source artifacts	Duplicate audit JSON, standardized final report, model/GUI/explainability scripts	Primary source for leakage counts, Phase 1/material-clean tables, architecture and explainability protocol
Tier C - current project status decision	User-confirmed roadmap and presentation conventions as of 2 Oct 2026	Authority for what is complete, pending, intentionally deferred, or presentation-only

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1. Executive summary

The project implements and extends a 2026 hybrid brain-tumor MRI classification pipeline using five pretrained CNN backbones connected to an LSTM and a downstream PCA + linear SVM classifier. The four classes are glioma, meningioma, no tumor and pituitary. The five locked backbones are VGG16, VGG19, ResNet50, ResNet152 and EfficientNet-B3.

Current stage
Phase 1 baseline, neural-head evaluation, PCA+SVM comparison, duplicate/leakage audit, material-clean evaluation, Phase 2A fixed-30-epoch training, Phase 2B hyperparameter-sensitivity training and analysis, GUI, Grad-CAM and final-SVM occlusion are complete. Phase 2C external validation is the next substantive experiment. Duplicate-group-aware re-splitting/retraining remains an unresolved stronger-remediation decision.

Key numerical findings

- Phase 1 standardized PCA+SVM test accuracy ranged from 93.52% (VGG19) to 98.47% (EfficientNet-B3); ResNet152 was 98.40%. The EfficientNet-B3 vs ResNet152 difference was only 0.07 percentage points and should not be exaggerated.
- The post-hoc material_clean test subset retained 1,061/1,311 test images after excluding 250 implicated test images. On this subset EfficientNet-B3 was 98.680% accurate (macro-F1 98.511%) and ResNet152 was 98.398% (macro-F1 98.315%).
- The leakage audit found 464 exact cross-split duplicate pairs plus 98 high-confidence near-duplicate pairs = 562 material cross-split pairs across 7,023 images. Patient-level independence cannot be established without reliable patient identifiers.
- Phase 2A + 2B provides a matched 40-run, fixed-30-epoch training set. All runs use seed 42, batch size 8, and the same saved split. Phase 2B comprises 35 one-factor perturbations.
- Learning rate was the dominant sensitivity dimension: mean within-backbone range in best validation accuracy was 4.62 pp, versus 0.58 pp for LSTM width and 0.42 pp for dropout.
- LSTM 512 produced a small increase in peak validation accuracy relative to LSTM 256 for all five backbones, but the gains were only +0.09 to +0.70 pp and did not always persist to epoch 30.
- The clearest unstable setting was ResNet50 at LR 5e-4: best validation 94.75% at epoch 25, falling to 90.99% by epoch 30 (-3.76 pp).
- These Phase 2 findings are validation-split sensitivity results, not new clean-test performance claims. They are single-seed experiments and the validation split is also affected by the known cross-split duplication issue.

Workstream	Status	Meaning
Phase 1 - five-backbone baseline	COMPLETE	Standardized neural + PCA/SVM pipeline
Neural-head test evaluation	COMPLETE	All five best checkpoints evaluated
Neural-head vs PCA+SVM comparison	COMPLETE	Correct term: downstream-classifier comparison, not PCA-only ablation
Duplicate/leakage audit	COMPLETE	MATERIAL_DUPLICATION; 562 material cross-split pairs
material_clean post-hoc test evaluation	COMPLETE	n=1,061; excludes 250 implicated test images
Phase 2A fixed 30 epochs	COMPLETE	5 matched baseline runs
Phase 2B sensitivity training	COMPLETE	35/35 variants available
Phase 2B integrity + analysis	COMPLETE	40-run matched analysis completed in this report
GUI / Grad-CAM / occlusion	COMPLETE	Research/educational prototype
Phase 2C external validation	NEXT	Frozen pipeline; no external tuning/refitting
Duplicate-group-aware re-split + retrain	UNRESOLVED	Scientifically stronger remediation; not yet performed

2. Project objective and locked scope

The original objective was a faithful implementation of the base-paper concept before adding extensions. The project then evolved into a reproducible five-backbone comparison with deployment-style inference, leakage auditing, interpretability checks and controlled hyperparameter sensitivity experiments.

Locked classification scope

- Input: ordinary MRI image files (PNG/JPG/JPEG/BMP/TIFF in the GUI); DICOM is intentionally outside the current scope.
- Classes: glioma, meningioma, notumor, pituitary.
- Backbones: VGG16, VGG19, ResNet50, ResNet152, EfficientNet-B3. The final project should be described as five backbones.
- Core architecture: MRI -> CNN spatial feature map -> spatial sequence -> LSTM -> BatchNorm -> Dropout -> Dense128/ReLU -> neural classifier head; the 128-D features also feed StandardScaler -> PCA(64) -> linear SVM.
- Final standardized prediction path uses the downstream SVM. The neural head remains important for baseline comparison and Grad-CAM.
- Softmax is inference-only for interpreting neural logits; it was not an extra training layer.

3. Dataset, preprocessing and architecture

3.1 Dataset and saved split

Partition	Images	Role
Original Kaggle Training folder	5,712	Source of fixed train/validation split
Saved training subset	4,569	Optimization and fitting of scaler/PCA/SVM
Saved validation subset	1,143	Checkpoint/model-selection and Phase 2 sensitivity
Official Kaggle Testing folder	1,311	Historical standardized test evaluation
material_clean test subset	1,061	Post-hoc leakage-aware test evaluation

Training augmentation is used only during neural-network training. Validation, test, post-training neural evaluation and deep-feature extraction use deterministic evaluation preprocessing. StandardScaler and PCA are fitted on training features only; validation/test are transformed using those fitted objects.

3.2 Neural architecture

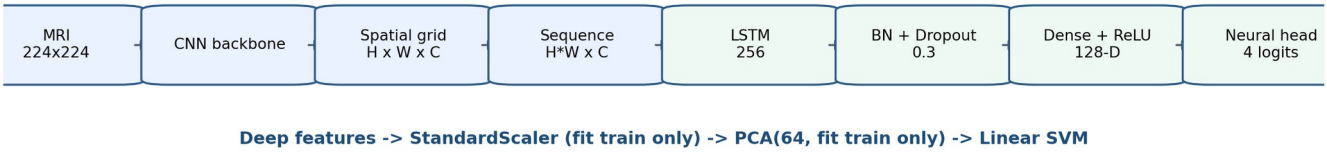


Figure 1. Locked CNN-LSTM architecture and downstream PCA+SVM path.

- LSTM hidden size baseline: 256; dense feature size: 128; dropout baseline: 0.3; 4 output logits.
- Training baseline: RMSprop, learning rate 1e-4, batch size 8, CrossEntropyLoss applied directly to raw logits.
- VGG16 and VGG19 convolutional backbones are frozen for transfer-learning stability after fully unfrozen VGG training collapsed. ResNet50/ResNet152/EfficientNet-B3 were handled as the fine-tuned family, so this training-policy asymmetry must be acknowledged.
- The spatial CNN grid is reshaped into a sequence rather than globally pooled before the LSTM. For ResNet152, for example, 7x7 spatial locations become a 49-step sequence with 2,048 features per position.

4. Project journey and methodology evolution

Milestone	What changed / why
Base-paper selection	Hybrid CNN+LSTM+SVM paper selected as implementation target; faithful base first, extensions later.
GPU framework change	TensorFlow path abandoned after RTX 5070 / Blackwell PTX toolchain incompatibility; project moved to PyTorch under Ubuntu/WSL2.
ResNet152 prototype	Architecture tested in stages: CNN feature map, reshape to spatial sequence, LSTM, post-LSTM block, complete forward/training step, then full training.
Generic five-backbone framework	Unified CNNLSTM model implemented for VGG16, VGG19, ResNet50, ResNet152 and EfficientNet-B3.
VGG stabilization	Fully unfrozen VGG behavior collapsed; pretrained VGG convolutional backbones were frozen for stable transfer learning.
Standardized evaluation	Each backbone trained/evaluated through a common deterministic feature-extraction -> StandardScaler -> PCA64 -> linear SVM path.
GUI + explainability	Streamlit GUI implemented; Grad-CAM added for the neural head, final-SVM occlusion added for the complete deployed path.
Leakage audit	Cross-split exact/perceptual audit revealed material duplication. Historical performance interpretation was revised accordingly.
Phase 2A	Five common fixed 30-epoch baseline runs completed without early termination for convergence/stability comparison.
Phase 2B	Seven one-factor hyperparameter variants per backbone completed (35 total) and compared against Phase 2A baselines.

5. Phase 1 baseline and downstream-classifier comparison

The correct terminology is neural-head vs PCA+SVM downstream-classifier comparison. It is not a PCA-only ablation because both dimensionality reduction and classifier family change together.

Backbone	Neural test acc.	Neural macro-F1	PCA+SVM test acc.	Acc. change
VGG16	93.67%	93.32%	94.97%	+1.30 pp
VGG19	93.14%	92.72%	93.52%	+0.38 pp
ResNet50	97.33%	97.19%	98.09%	+0.76 pp
ResNet152	98.17%	98.06%	98.40%	+0.23 pp
EfficientNet-B3	98.02%	97.94%	98.47%	+0.45 pp

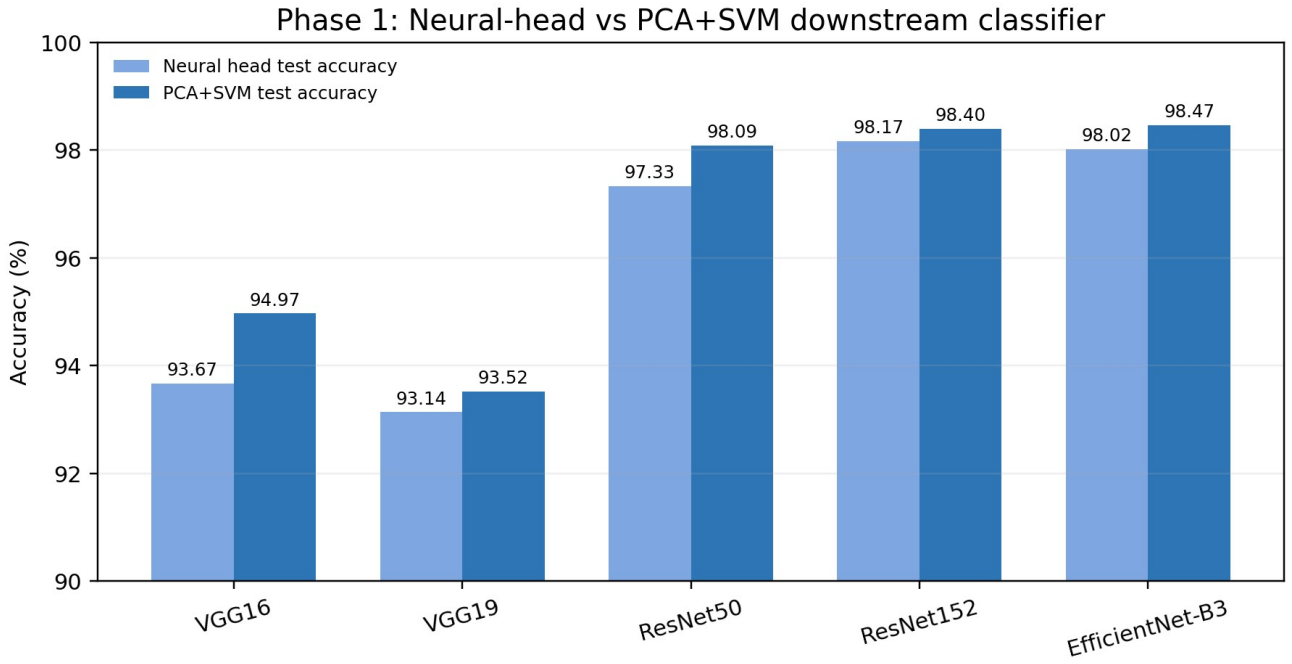


Figure 2. Phase 1 test accuracy: neural head vs downstream PCA+SVM classifier.

Interpretation rule
Test macro-F1 also increased for all five backbones, but validation did not improve uniformly. Therefore do not claim that PCA+SVM universally improves performance across every split. Also do not overstate the 0.07 pp EfficientNet-B3 vs ResNet152 test-accuracy difference.

6. Leakage audit and material-clean evaluation

6.1 Cross-split duplicate/leakage audit

Audit item	Result
Images audited	7,023
Exact cross-split duplicate pairs	464
High-confidence near-duplicate pairs	98
Material cross-split pairs	562
Additional perceptual review candidates	481
Unique official-test images implicated	250
material_clean test size	1,061 of 1,311
Patient-level independence	Cannot be established without reliable patient identifiers

The audit used exact file/pixel hashing plus perceptual candidate screening. The recorded thresholds include pHash Hamming <=4 and dHash Hamming <=6 for candidate generation, with grayscale MAE/correlation checks for perceptual review. The result status is MATERIAL_DUPLICATION.

What the audit changes
The original validation/test numbers remain real recorded outputs, but they are not clean independent-generalization estimates. This caveat applies to the Phase 2 validation sensitivity study as well because it uses the same saved

split. The material_clean evaluation reduces direct test-image contamination post hoc, but it does not undo training on a split containing duplicated groups and it cannot prove patient-level independence.

6.2 Phase 1 material-clean test results

Backbone	material_clean accuracy	material_clean macro-F1	Original test accuracy
VGG16	94.816%	94.302%	94.966%
VGG19	93.025%	92.415%	93.516%
ResNet50	98.303%	98.134%	98.093%
ResNet152	98.398%	98.315%	98.398%
EfficientNet-B3	98.680%	98.511%	98.474%

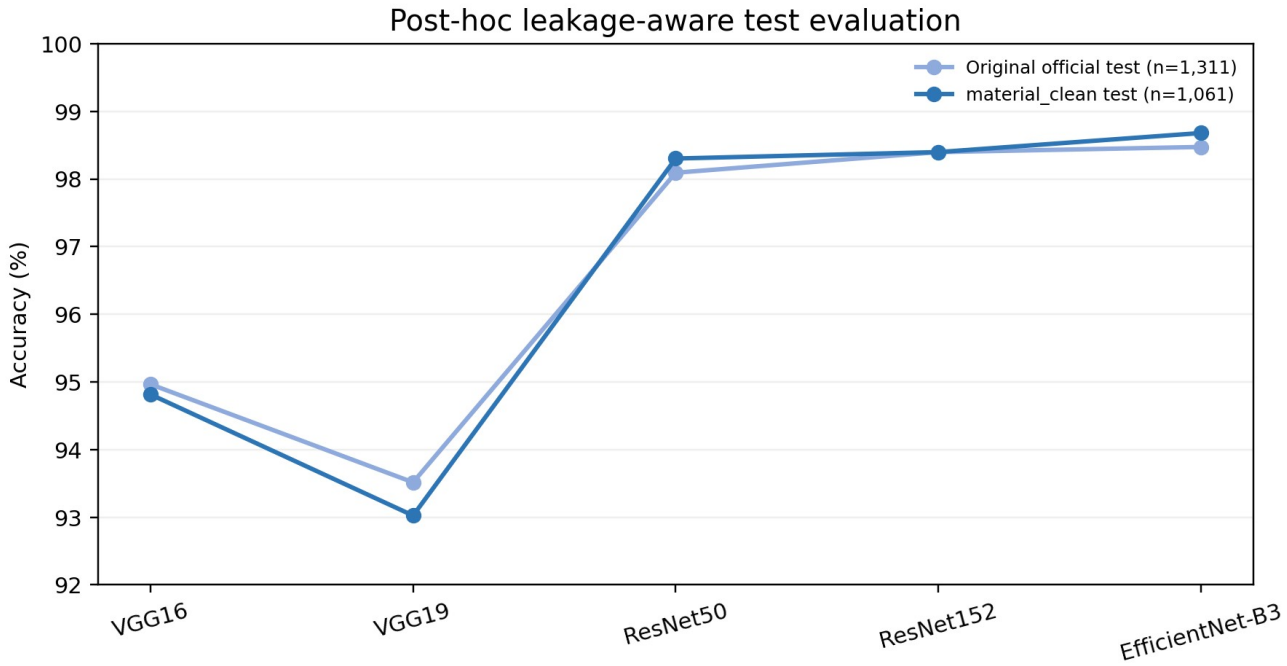


Figure 3. Historical standardized test accuracy versus post-hoc material_clean accuracy.

Removing implicated test images did not collapse performance, but the result must still be described narrowly: it is a post-hoc leakage-aware subset evaluation, not a replacement for duplicate-group-aware re-splitting and complete retraining.

7. GUI and explainability work

7.1 Streamlit GUI

- Select any of the five standardized backbones and upload an MRI image.
- Display final SVM prediction, neural-head comparison, agreement status, scores, runtime and inference details.
- Display validation/test metrics, confusion matrices, per-class results, training curves and cross-backbone comparison.
- Load saved scaler/PCA/SVM artifacts; nothing is refitted on an uploaded image.
- Neural-head softmax is an inference view of logits, not a training layer. SVM decision values are shown as relative scores, not calibrated clinical probabilities.
- The interface is explicitly a research/educational prototype, not a medical device.

7.2 Grad-CAM evaluation

Formal Grad-CAM evaluation was performed on ResNet152 using a fixed seed-42 sample of 20 official-test images (5 per class). ResNet Layer 3, Layer 4 and a 50:50 fused map were reviewed. Among 15 tumor images:

Map	Good	Partial	Poor	Chosen most relevant
Layer 3 (14x14)	1	9	5	9/15
Layer 4 (7x7)	3	2	10	6/15
Fused 50:50	2	6	7	0/15

- Neural and SVM heads agreed on all 20 sampled images and were both correct on 19/20. The shared error was pituitary Te-pi_0078.jpg, predicted as glioma.
- Correct classification frequently coexisted with poor or off-lesion localization; hotspots often involved skull margin, face, image border or extracranial regions.
- Grad-CAM therefore must be described as a coarse qualitative contribution map, not a tumor-segmentation boundary or proof of clinically valid reasoning.

7.3 Final-SVM occlusion comparison

- 11 fixed cases selected before occlusion review; 9 tumor + 2 no-tumor controls.
- Patch 32x32, stride 16, 169 occlusions/image; the complete CNN-LSTM-StandardScaler-PCA-SVM path is rerun.
- Among 9 tumor cases, occlusion localization was Good 2, Partial 2, Poor 5. Grad-CAM was judged more anatomically relevant in 4 cases, occlusion in 3, and both were predominantly off-lesion in 2.
- Across 1,859 occlusions there were 58 class flips (3.12% pooled); median per-image flip rate was 0%; 5/11 images had any flip.
- Te-pi_0078.jpg accounted for 38/169 flips (22.5%), suggesting marked instability around the shared misclassification.
- Occlusion maps are independently normalized per image; vivid heatmaps can represent very small absolute score changes. Interpret maps together with numerical score drops and flip rates.

8. Phase 2A - common fixed 30-epoch study

Phase 2A removes early termination as a confounder and trains each baseline configuration for exactly 30 epochs. The goal is not to replace Phase 1 final-test evaluation; it is to compare convergence, late-epoch stability and overfitting behavior under a common training horizon.

Backbone	LR	LSTM	Dropout	Best ep.	Best val	Ep30 val	Ep30 train	Final gap
VGG16	1e-04	256	0.3	30	95.45%	95.45%	98.40%	2.95 pp
VGG19	1e-04	256	0.3	30	95.19%	95.19%	97.88%	2.69 pp
ResNet50	1e-04	256	0.3	30	98.43%	98.43%	98.99%	0.57 pp
ResNet152	1e-04	256	0.3	10	98.78%	98.69%	99.32%	0.63 pp
EfficientNet-B3	1e-04	256	0.3	27	99.30%	98.69%	99.56%	0.87 pp

Baseline observations

VGG16 and VGG19 continued improving through epoch 30. ResNet152 peaked at epoch 10 (98.78%) and remained close at epoch 30 (98.69%). EfficientNet-B3 peaked at epoch 27 (99.30%) and ended at 98.69%. These trajectories motivated the explicit Phase 2B sensitivity study.

9. Phase 2B - hyperparameter-sensitivity study

9.1 Design and integrity check

Phase 2B uses a one-factor-at-a-time design around the fixed baseline LR 1e-4 / LSTM 256 / dropout 0.3. Seven non-baseline variants were run for each of five backbones = 35 sensitivity runs. The five Phase 2A baselines are added only for comparison, creating a 40-run analysis set.

Variant	Learning rate	LSTM hidden	Dropout
LR low	1e-5	256	0.3
LR mid-low	5e-5	256	0.3
LR high	5e-4	256	0.3
LSTM small	1e-4	128	0.3
LSTM large	1e-4	512	0.3
Dropout low	1e-4	256	0.2
Dropout high	1e-4	256	0.5
Reference baseline	1e-4	256	0.3

- All 40 JSON files report epochs_requested=30 and epochs_completed=30.
- All runs use seed 42 and batch size 8; Dense128 remains fixed.
- For every run, stored best_val_accuracy matches the maximum value in the 30-epoch validation-accuracy history.
- The queue completed 34 newly executed runs and skipped the already-complete VGG16 LR 1e-5 run. All 35 planned sensitivity configurations are therefore available.

9.2 Strongest Phase 2B perturbation per backbone

Backbone	Strongest perturbation	Baseline best	Variant best	Delta	Best ep.	Ep30 val	Approx. val-image delta
VGG16	lr_5e-4	95.45%	95.98%	+0.52 pp	27	95.98%	~+6 images
VGG19	lstm_512	95.19%	95.89%	+0.70 pp	30	95.89%	~+8 images
ResNet50	dropout_0.2	98.43%	99.13%	+0.70 pp	28	98.51%	~+8 images
ResNet152	lstm_512	98.78%	99.21%	+0.44 pp	26	98.78%	~+5 images
EfficientNet-B3	lstm_512	99.30%	99.39%	+0.09 pp	29	98.43%	~+1 images

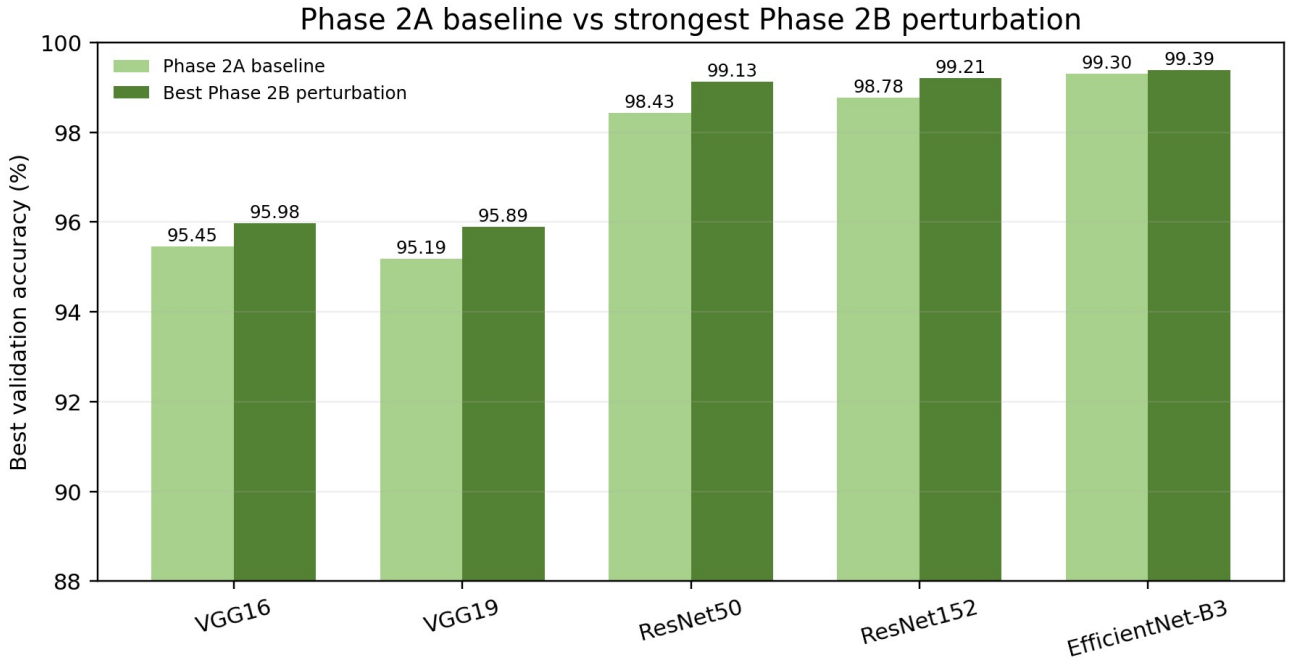


Figure 4. Best fixed-30-epoch validation accuracy: Phase 2A baseline vs strongest Phase 2B perturbation.

Do not overclaim the small gains

With 1,143 validation images, one image corresponds to about 0.0875 percentage points. The strongest gains are equivalent to roughly +1 to +8 validation images, and all Phase 2 runs use a single seed. These are useful sensitivity signals, not evidence of statistically established superiority.

9.3 Learning rate is the dominant tested sensitivity

Hyperparameter	VGG16	VGG19	ResNet50	ResNet152	EffNet-B3	Mean range (pp)
Learning rate	5.77	5.60	3.85	4.99	2.89	4.62
LSTM width	0.44	1.22	0.61	0.44	0.17	0.58
Dropout	0.17	0.52	0.79	0.26	0.35	0.42

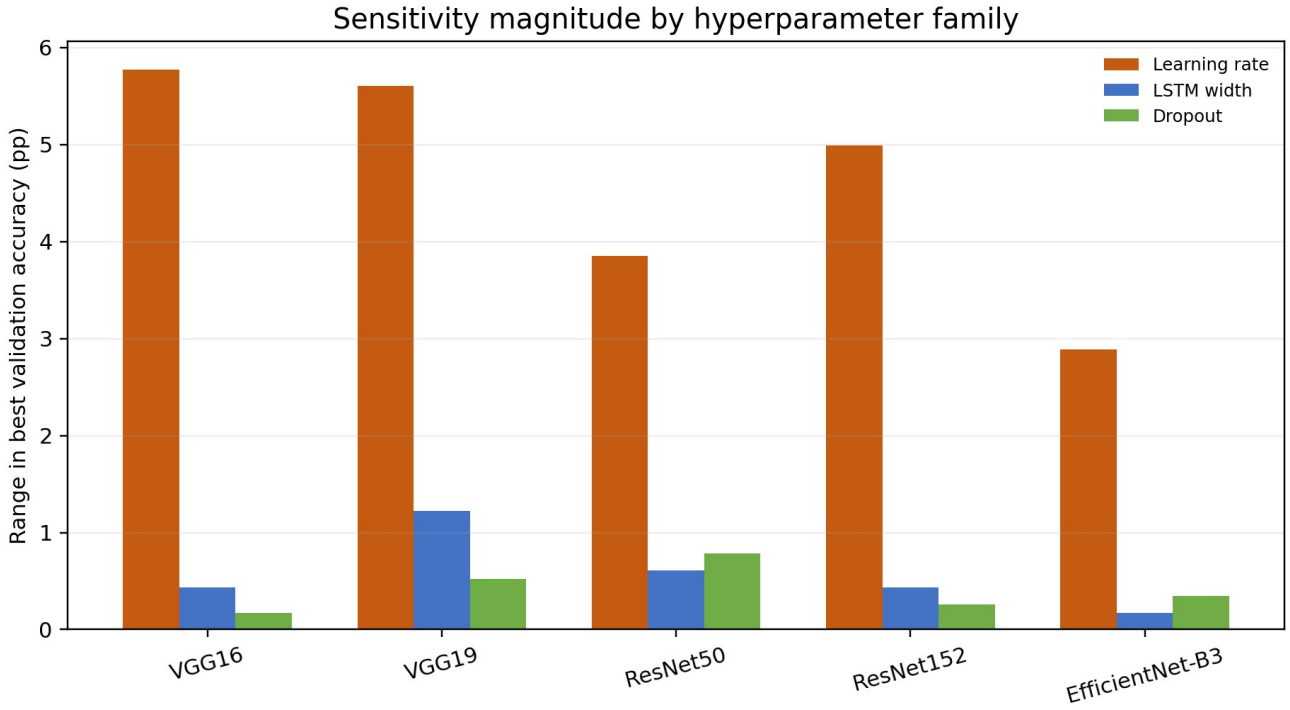


Figure 5. Within-backbone range in peak validation accuracy across each tested hyperparameter family.

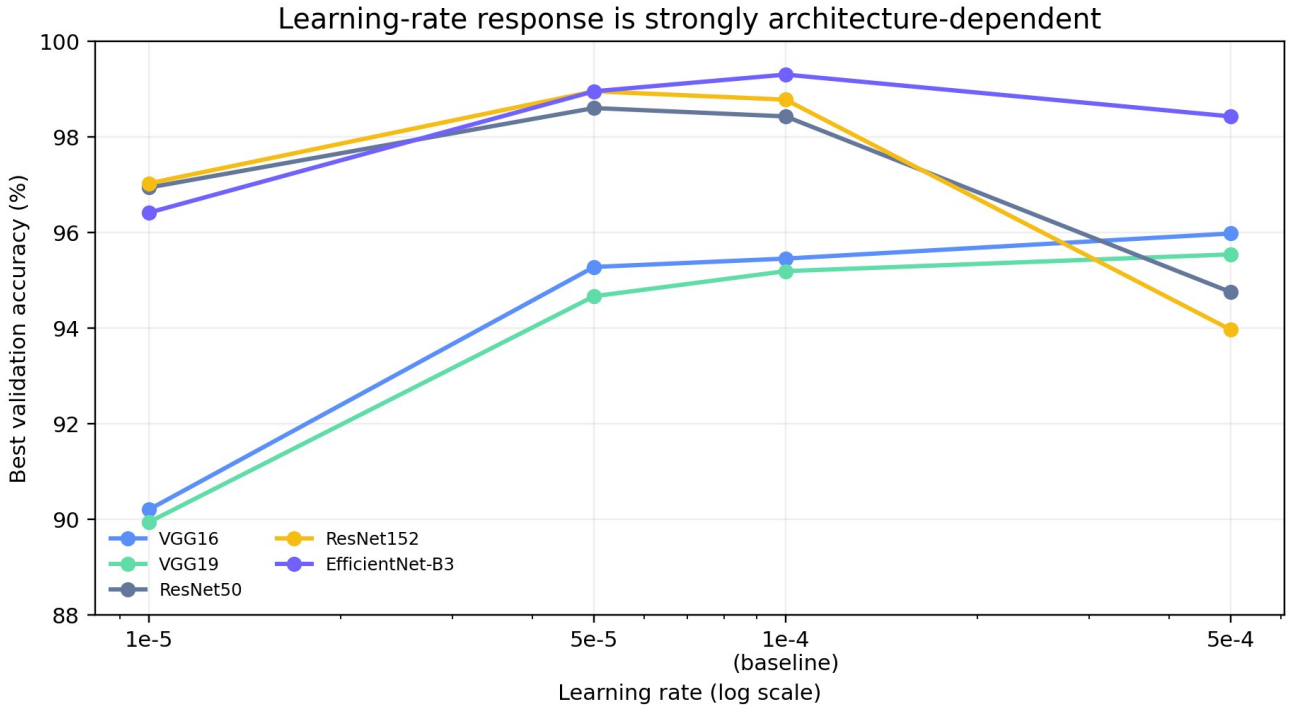


Figure 6. Learning-rate response across the five backbones.

- VGG16/VGG19: LR 1e-5 was strongly underpowered (-5.25 pp vs baseline for both); LR 5e-4 was comparatively well tolerated and slightly exceeded baseline peak validation.
- ResNet50/ResNet152: LR 5e-4 was damaging (-3.67 pp and -4.81 pp vs baseline respectively); 5e-5 was slightly above baseline peak validation.
- EfficientNet-B3: baseline 1e-4 remained the best of the tested LR-only settings; 1e-5 reduced peak validation by 2.89 pp.
- Therefore the supported conclusion is architecture-dependent LR sensitivity, not a universal recommendation for higher or lower LR.

9.4 LSTM width: modest but unusually consistent peak improvement at 512

Backbone	LSTM128 peak	LSTM256 baseline	LSTM512 peak	512 delta	512 best ep.	512 ep30
VGG16	95.80%	95.45%	95.89%	+0.44 pp	19	95.63%
VGG19	94.66%	95.19%	95.89%	+0.70 pp	30	95.89%
ResNet50	98.43%	98.43%	99.04%	+0.61 pp	29	98.25%
ResNet152	98.95%	98.78%	99.21%	+0.44 pp	26	98.78%
EfficientNet-B3	99.21%	99.30%	99.39%	+0.09 pp	29	98.43%

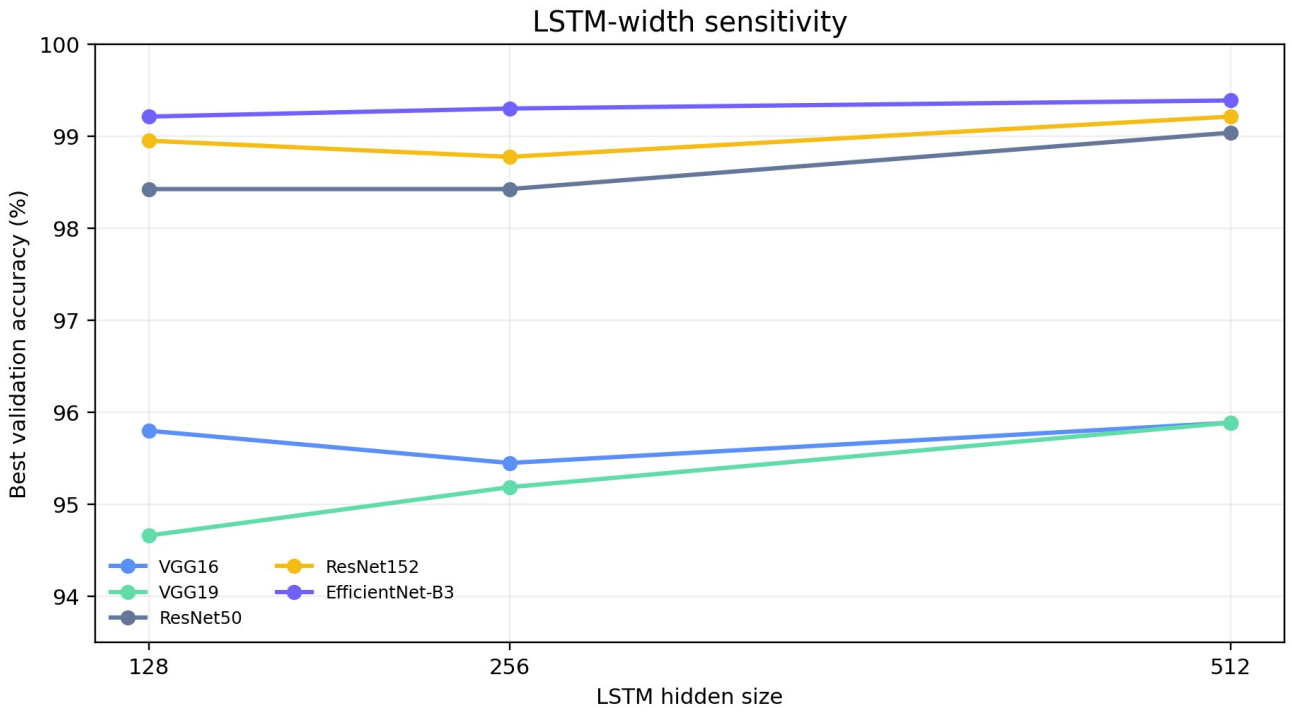


Figure 7. Peak validation accuracy across LSTM widths 128, 256 and 512.

LSTM512 raised peak validation accuracy relative to baseline LSTM256 for all five backbones (+0.09 to +0.70 pp; mean about +0.45 pp). However, the peak did not always persist: EfficientNet-B3 LSTM512 peaked at 99.39% at epoch 29 and finished epoch 30 at 98.43%. The safest wording is that LSTM512 was associated with modest peak improvements, not that it was universally superior or more stable.

9.5 Dropout: comparatively small and inconsistent effects

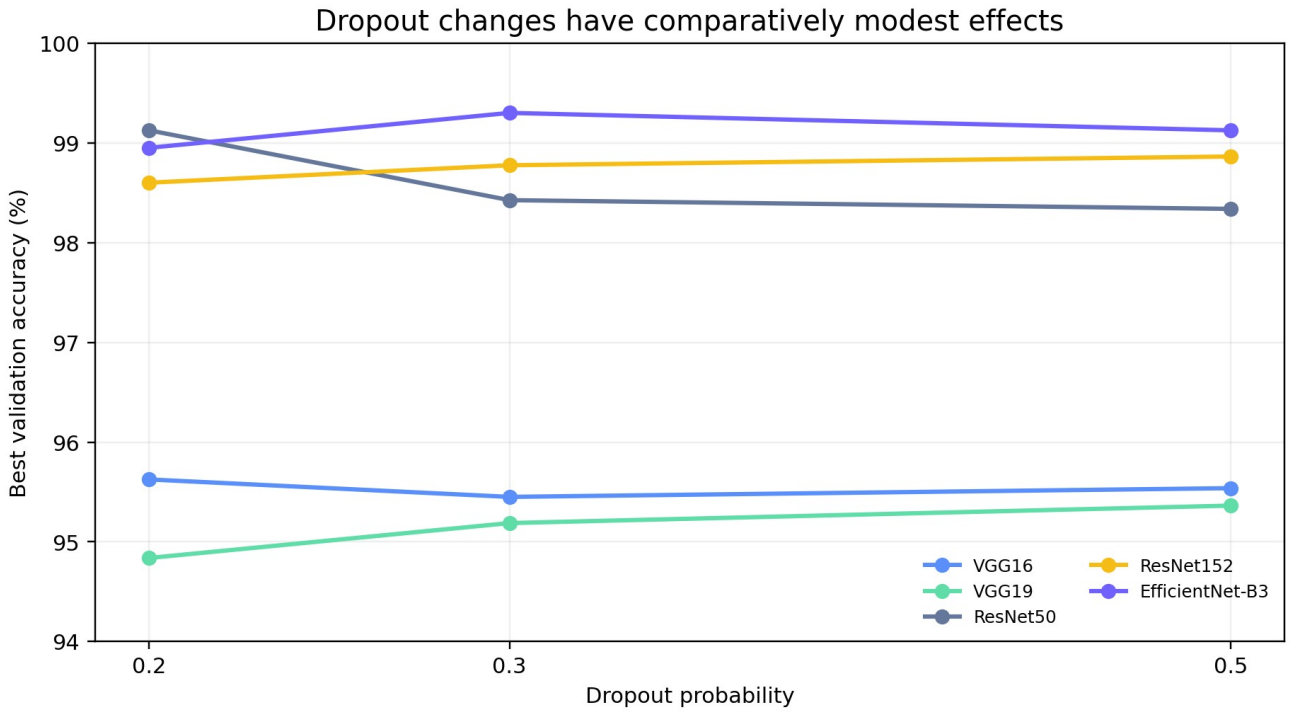


Figure 8. Peak validation accuracy across dropout 0.2, 0.3 baseline and 0.5.

The mean within-backbone range across dropout settings was only 0.42 pp. No single value was preferred by all architectures. ResNet50 peaked with dropout 0.2 (+0.70 pp vs baseline), while EfficientNet-B3 baseline dropout 0.3 remained higher than either 0.2 or 0.5.

9.6 Late-epoch stability

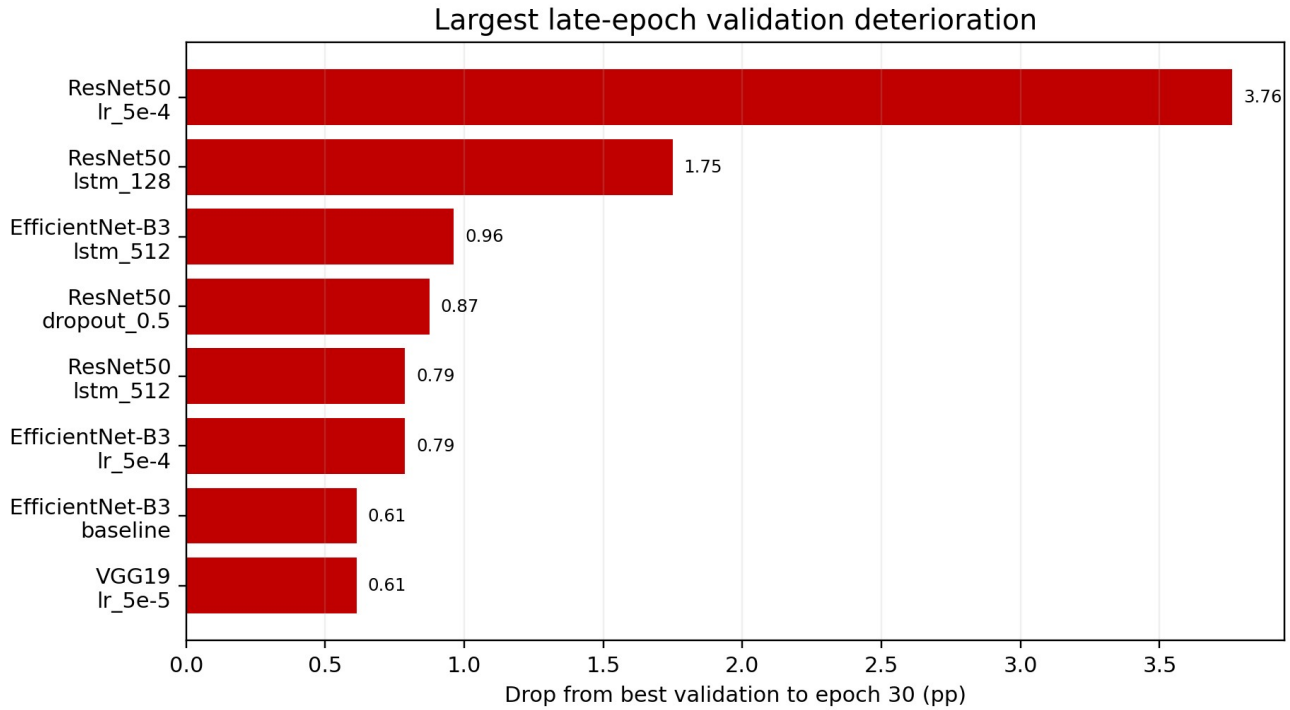


Figure 9. Largest decreases from each run's best validation accuracy to epoch 30.

The clearest instability was ResNet50 LR 5e-4: peak validation 94.75% at epoch 25 fell to 90.99% at epoch 30 (-3.76 pp), while epoch-30 train accuracy remained 95.16%, producing a 4.17 pp train-validation gap. This setting is a clear example of why fixed-horizon curves reveal information that a best-checkpoint-only summary would hide.

10. Integrated Phase 2 findings

ID	Finding	Evidence-based interpretation
F1	Learning rate dominated sensitivity	Mean peak-validation range 4.62 pp vs 0.58 pp LSTM and 0.42 pp dropout.
F2	LR preference is architecture-dependent	Frozen VGGs tolerated 5e-4 relatively well; fine-tuned ResNets degraded strongly at 5e-4; EfficientNet-B3 baseline 1e-4 remained strongest among LR-only settings.
F3	LSTM512 gave the most consistent positive peak shift	All five backbones improved in peak validation vs LSTM256 baseline, but by only +0.09 to +0.70 pp.
F4	Peak accuracy and stability are different questions	Some high peaks were followed by late-epoch decline, especially ResNet50 LR 5e-4 and EfficientNet-B3 LSTM512.
F5	Dropout was a secondary factor in the tested range	0.2-0.5 produced relatively small, inconsistent changes.
F6	High baseline performance leaves little headroom	Best perturbation gains often represented only a few validation images.
F7	Phase 2 does not establish external generalization	All runs use one saved validation split, one seed, and a dataset known to contain cross-split duplication.

Report-ready Phase 2 conclusion

Across five CNN-LSTM architectures, learning rate produced substantially greater variation in validation performance than the tested changes in LSTM hidden size or dropout. Learning-rate sensitivity was architecture-dependent. Increasing LSTM hidden size to 512 produced small increases in peak validation accuracy across all five backbones, although these gains did not always persist to epoch 30. Dropout variation between 0.2 and 0.5 had comparatively modest and inconsistent effects. Because the experiments used a single saved validation split and one random seed, the findings should be interpreted as a controlled sensitivity analysis rather than proof of a new optimal configuration.

11. Scientific limitations and interpretation rules

Limitation	Required interpretation
Cross-split duplication	The saved split is not image-level independent. Phase 1 and Phase 2 validation/test metrics can be optimistic.
Patient identity unavailable	The audit cannot establish patient-level separation or patient-level leakage.
Single seed	Phase 2A/2B use seed 42 only. No variance across random initializations is available.
Validation resolution	n=1,143 means one image = ~0.0875 pp; several observed gains are only a handful of cases.
No external validation yet	Performance on an independent source population/scanner/site is unknown.

Limitation	Required interpretation
VGG training asymmetry	VGG backbones are frozen while residual/EfficientNet models use a different fine-tuning policy; comparisons are practical pipeline comparisons, not perfectly symmetric optimization experiments.
Hyperparameter coverage	Only LR, LSTM width and dropout were tested, one factor at a time. Interactions were not explored.
No statistical significance claim	The project has not established confidence intervals or repeated-seed significance for Phase 2 differences.
Explainability is qualitative	Grad-CAM explains the neural head; occlusion explains the final SVM. Neither is tumor segmentation or proof of clinically valid localization.
Prototype status	The system is research/educational and must not be presented as a diagnostic medical device.

Terminology that must remain consistent

- Use "neural-head vs PCA+SVM downstream-classifier comparison"; do not call it a PCA-only ablation.
- Use "material_clean post-hoc test subset"; do not imply it creates a newly independent dataset.
- Describe original standardized metrics as historical contaminated-split results after the audit.
- Describe Phase 2 best values as validation accuracy, not final test accuracy.
- Say LSTM512 produced modest peak improvements; do not say it is universally best or statistically superior.
- Do not claim universal PCA+SVM improvement on validation; the universal improvement statement applies only to test accuracy/macro-F1 in the completed Phase 1 comparison.
- Do not call Grad-CAM or occlusion a tumor boundary, segmentation mask, or evidence of clinical trustworthiness.
- Do not exaggerate EfficientNet-B3 vs ResNet152; Phase 1 standardized test accuracy differed by only 0.07 pp.

12. Current project state and next experiments

12.1 Immediate next phase - Phase 2C external validation

1. Select an independent external dataset and document provenance, acquisition context and licensing.
2. Perform an overlap audit against the current Kaggle corpus before evaluation.
3. Define and document class mapping to glioma / meningioma / notumor / pituitary; exclude unsupported classes transparently.
4. Use the same deterministic evaluation preprocessing. Do not augment external evaluation images.
5. Freeze the selected trained CNN-LSTM checkpoint, StandardScaler, PCA and SVM. Do not tune, refit or recalibrate on the external dataset.
6. Report accuracy, macro precision/recall/F1, per-class metrics, confusion matrix and distribution-shift observations.
7. Keep external validation separate from hyperparameter selection; it is an out-of-source evaluation, not another tuning set.

12.2 Unresolved stronger remediation - duplicate-group-aware re-split/retraining

The scientifically stronger response to the leakage audit is to construct duplicate/near-duplicate groups, split groups rather than individual images, retrain all five backbones from scratch, and refit each scaler/PCA/SVM using the new training split only. This has not been performed. The current project roadmap treats it as an important unresolved decision after Phase 2C, not as completed work.

If the project makes a clean-generalization claim

A duplicate-group-aware re-split + complete retraining is the appropriate stronger remediation before treating internal test performance as independent evidence. Even then, patient-level independence cannot be proven without reliable patient identifiers.

12.3 Presentation scope note

Review-I PPT intentionally lags the actual project. Its main narrative may stop at CNN-LSTM and the quantified neural-head baseline, leaving PCA+SVM as "next" in the main storyline and placing the actual downstream-classifier comparison in backup/appendix. This is a presentation choice, not the true implementation status.

13. Claude / future-chat transfer instructions

Authority rule
Treat this document as the project-status snapshot as of 2 October 2026. If an older report, PPT or conversation conflicts with it, prefer the newer verified Phase 2 result packages and the current status decisions recorded here. Do not silently "correct" the project into a different methodology.

13.1 What Claude should preserve

- Five-backbone final scope only: VGG16, VGG19, ResNet50, ResNet152, EfficientNet-B3.
- Architecture: MRI -> CNN -> spatial sequence -> LSTM(256 baseline) -> BN -> Dropout 0.3 -> Dense128/ReLU -> neural classifier; 128-D features -> StandardScaler -> PCA64 -> linear SVM.
- RMSprop, LR 1e-4 baseline, batch 8, raw-logit CrossEntropyLoss, training-only augmentation, deterministic evaluation.
- VGG backbones frozen for stability; acknowledge this asymmetry.
- Duplicate audit is already complete; it is not pending.
- Phase 2A and Phase 2B are complete; Phase 2C is next.
- Phase 2B has 35/35 sensitivity variants plus 5 Phase 2A baselines = 40 matched fixed-30-epoch runs.
- Learning-rate sensitivity is the strongest Phase 2 finding; LSTM512 gives modest peak improvements; dropout effects are small/inconsistent.
- Original split metrics must carry the leakage caveat; material_clean is post-hoc, not proof of independence.
- Grad-CAM explains the neural head; occlusion explains the final SVM; neither is segmentation.
- Claude is used for PPT/document generation/editing. ChatGPT handles code, experiments, methodology, result auditing and scientific interpretation.

13.2 Mistakes to avoid in future documents

- Saying the duplicate/leakage audit is still pending.
- Saying Phase 2B is incomplete or 1/35 complete.
- Calling the comparison "PCA ablation".
- Using a Phase 2 validation peak as a new test-set result.
- Claiming LSTM512 is proven best or statistically significant.
- Claiming dropout 0.2 or 0.5 is universally optimal.
- Claiming external validation has been performed.
- Claiming patient-level independence.
- Calling the GUI clinically deployable or the saliency maps tumor segmentation.
- Allowing Review-I presentation scope to overwrite the true current project state.

13.3 Compact handoff block

```
PROJECT: DL TUMOR CLASSIFICATION. STATUS 2026-10-02. Five-backbone Phase 1 is complete (VGG16, VGG19, ResNet50, ResNet152,
EfficientNet-B3). Locked architecture: MRI -> CNN -> spatial sequence -> LSTM -> BN -> Dropout -> Dense128 -> neural head;
Dense128 features -> StandardScaler -> PCA64 -> linear SVM. Baseline training uses RMSprop, LR 1e-4, batch 8, raw-logit
CrossEntropyLoss; augmentation train-only; eval deterministic; VGG16/19 frozen. Neural-head test accuracies: 93.67, 93.14,
97.33, 98.17, 98.02%. PCA+SVM test accuracies: 94.97, 93.52, 98.09, 98.40, 98.47%. Duplicate audit complete: 7,023 images, 464
exact cross-split pairs, 98 high-confidence near-duplicate pairs, 562 material pairs; 250 test images implicated;
material_clean n=1,061; patient independence unknown. material_clean Phase 1 accuracies: 94.816, 93.025, 98.303, 98.398,
98.680%. Phase 2A complete: five fixed 30-epoch baselines. Phase 2B complete: 35/35 one-factor sensitivity runs; full 40-run
analysis complete. Main Phase 2 finding: mean peak-validation sensitivity range LR 4.62 pp, LSTM width 0.58 pp, dropout 0.42
pp. LSTM512 modestly increased peak validation for all five, but not always epoch-30 stability. Phase 2C external validation
is next. Duplicate-group-aware re-split/retraining remains unresolved. GUI, Grad-CAM and SVM occlusion are complete;
explainability is qualitative only. Review-I intentionally lags actual implementation.
```

Appendix A. Full 40-run Phase 2 result tables

All values below are derived directly from the uploaded training_results.json files. Delta is relative to the Phase 2A baseline peak validation accuracy for the same backbone. “Drift” = epoch-30 validation accuracy minus peak validation accuracy (negative values indicate late-epoch decline).

VGG16

Run	LR	LSTM	Drop	Best ep	Best val	Ep30 val	Delta pp	Drift pp
baseline	1e-04	256	0.3	30	95.451%	95.451%	+0.000	+0.000
lr_1e-5	1e-05	256	0.3	30	90.201%	90.201%	-5.249	+0.000
lr_5e-5	5e-05	256	0.3	30	95.276%	95.276%	-0.175	+0.000
lr_5e-4	5e-04	256	0.3	27	95.976%	95.976%	+0.525	+0.000
lstm_128	1e-04	128	0.3	30	95.801%	95.801%	+0.350	+0.000
lstm_512	1e-04	512	0.3	19	95.888%	95.626%	+0.437	-0.262
dropout_0.2	1e-04	256	0.2	27	95.626%	95.538%	+0.175	-0.087
dropout_0.5	1e-04	256	0.5	30	95.538%	95.538%	+0.087	+0.000

VGG19

Run	LR	LSTM	Drop	Best ep	Best val	Ep30 val	Delta pp	Drift pp
baseline	1e-04	256	0.3	30	95.188%	95.188%	+0.000	+0.000
lr_1e-5	1e-05	256	0.3	30	89.939%	89.939%	-5.249	+0.000
lr_5e-5	5e-05	256	0.3	28	94.663%	94.051%	-0.525	-0.612
lr_5e-4	5e-04	256	0.3	27	95.538%	95.013%	+0.350	-0.525
lstm_128	1e-04	128	0.3	28	94.663%	94.313%	-0.525	-0.350
lstm_512	1e-04	512	0.3	30	95.888%	95.888%	+0.700	+0.000
dropout_0.2	1e-04	256	0.2	29	94.838%	94.751%	-0.350	-0.087
dropout_0.5	1e-04	256	0.5	28	95.363%	95.188%	+0.175	-0.175

ResNet50

Run	LR	LSTM	Drop	Best ep	Best val	Ep30 val	Delta pp	Drift pp
baseline	1e-04	256	0.3	30	98.425%	98.425%	+0.000	+0.000
lr_1e-5	1e-05	256	0.3	27	96.938%	96.763%	-1.487	-0.175
lr_5e-5	5e-05	256	0.3	23	98.600%	98.425%	+0.175	-0.175
lr_5e-4	5e-04	256	0.3	25	94.751%	90.989%	-3.675	-3.762
lstm_128	1e-04	128	0.3	28	98.425%	96.675%	+0.000	-1.750
lstm_512	1e-04	512	0.3	29	99.038%	98.250%	+0.612	-0.787
dropout_0.2	1e-04	256	0.2	28	99.125%	98.513%	+0.700	-0.612
dropout_0.5	1e-04	256	0.5	26	98.338%	97.463%	-0.087	-0.875

ResNet152

Run	LR	LSTM	Drop	Best ep	Best val	Ep30 val	Delta pp	Drift pp
baseline	1e-04	256	0.3	10	98.775%	98.688%	+0.000	-0.087
lr_1e-5	1e-05	256	0.3	29	97.025%	96.938%	-1.750	-0.087
lr_5e-5	5e-05	256	0.3	17	98.950%	98.513%	+0.175	-0.437
lr_5e-4	5e-04	256	0.3	20	93.963%	93.613%	-4.812	-0.350
lstm_128	1e-04	128	0.3	29	98.950%	98.688%	+0.175	-0.262
lstm_512	1e-04	512	0.3	26	99.213%	98.775%	+0.437	-0.437
dropout_0.2	1e-04	256	0.2	27	98.600%	98.075%	-0.175	-0.525
dropout_0.5	1e-04	256	0.5	29	98.863%	98.425%	+0.087	-0.437

EfficientNet-B3

Run	LR	LSTM	Drop	Best ep	Best val	Ep30 val	Delta pp	Drift pp
baseline	1e-04	256	0.3	27	99.300%	98.688%	+0.000	-0.612
lr_1e-5	1e-05	256	0.3	28	96.413%	96.325%	-2.887	-0.087
lr_5e-5	5e-05	256	0.3	14	98.950%	98.950%	-0.350	+0.000
lr_5e-4	5e-04	256	0.3	27	98.425%	97.638%	-0.875	-0.787
lstm_128	1e-04	128	0.3	30	99.213%	99.213%	-0.087	+0.000
lstm_512	1e-04	512	0.3	29	99.388%	98.425%	+0.087	-0.962
dropout_0.2	1e-04	256	0.2	15	98.950%	98.688%	-0.350	-0.262
dropout_0.5	1e-04	256	0.5	26	99.125%	98.688%	-0.175	-0.437

Appendix B. Fixed-30-epoch validation trajectories

These plots are descriptive. They are useful for diagnosing convergence and instability; they should not be used to select a final model by visually inspecting the test set.

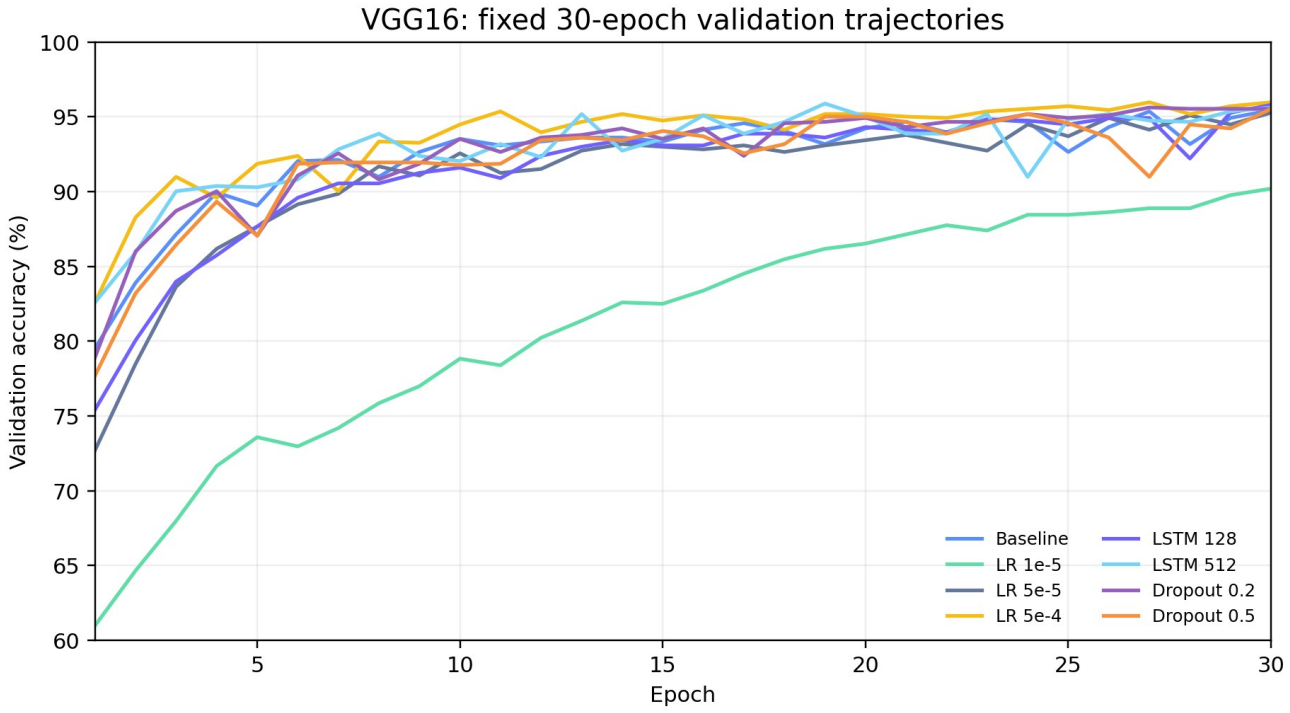


Figure B1. VGG16 validation accuracy across the baseline and seven Phase 2B perturbations.

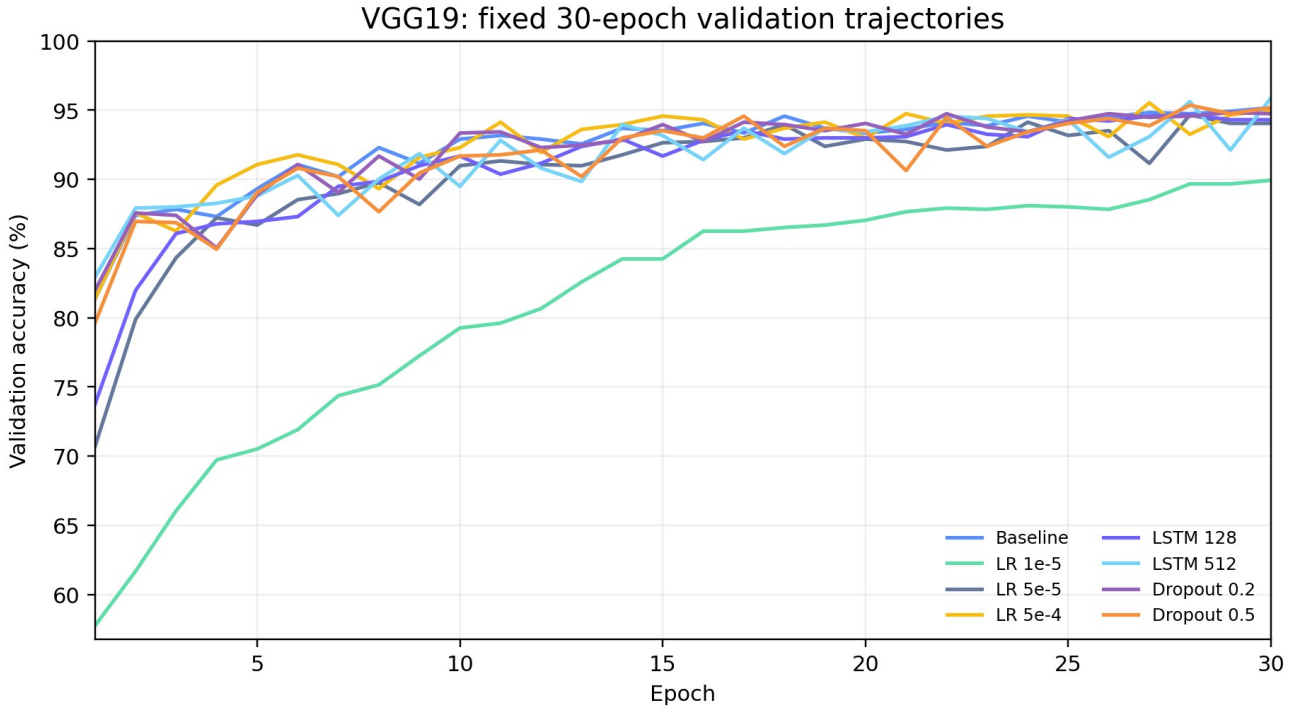


Figure B2. VGG19 validation accuracy across the baseline and seven Phase 2B perturbations.

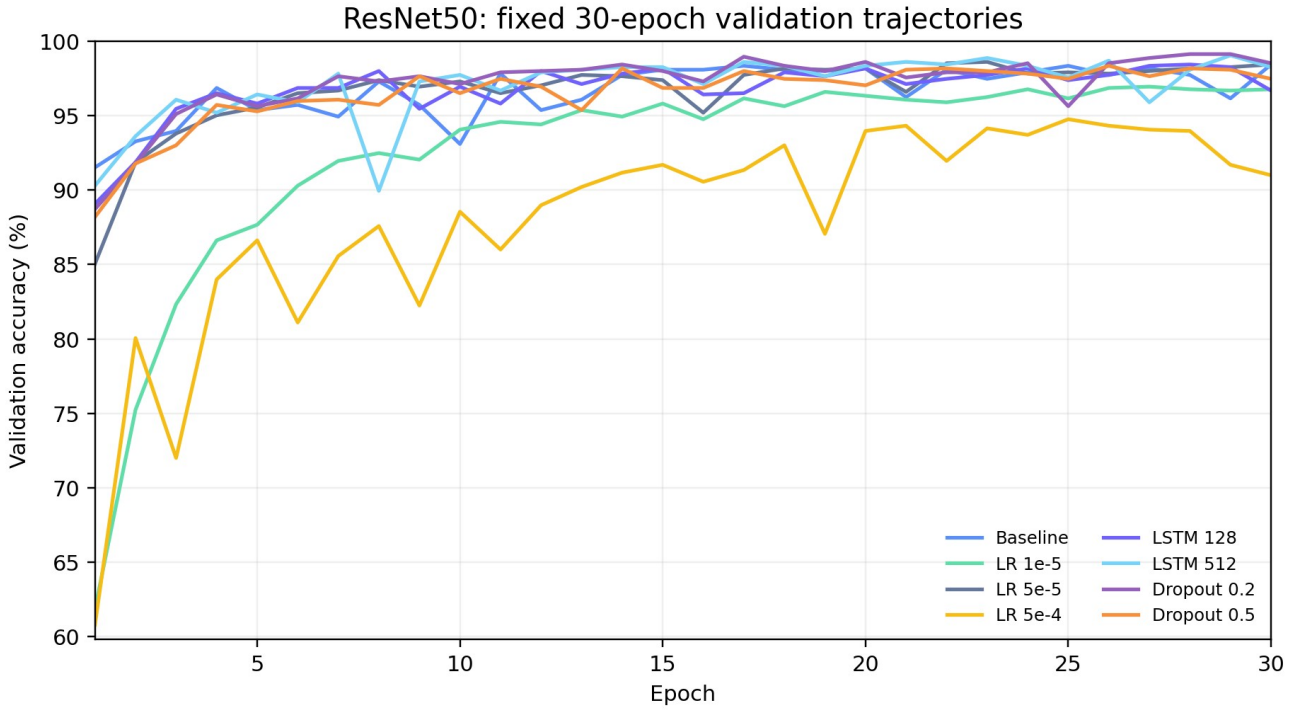


Figure B3. ResNet50 validation accuracy across the baseline and seven Phase 2B perturbations.

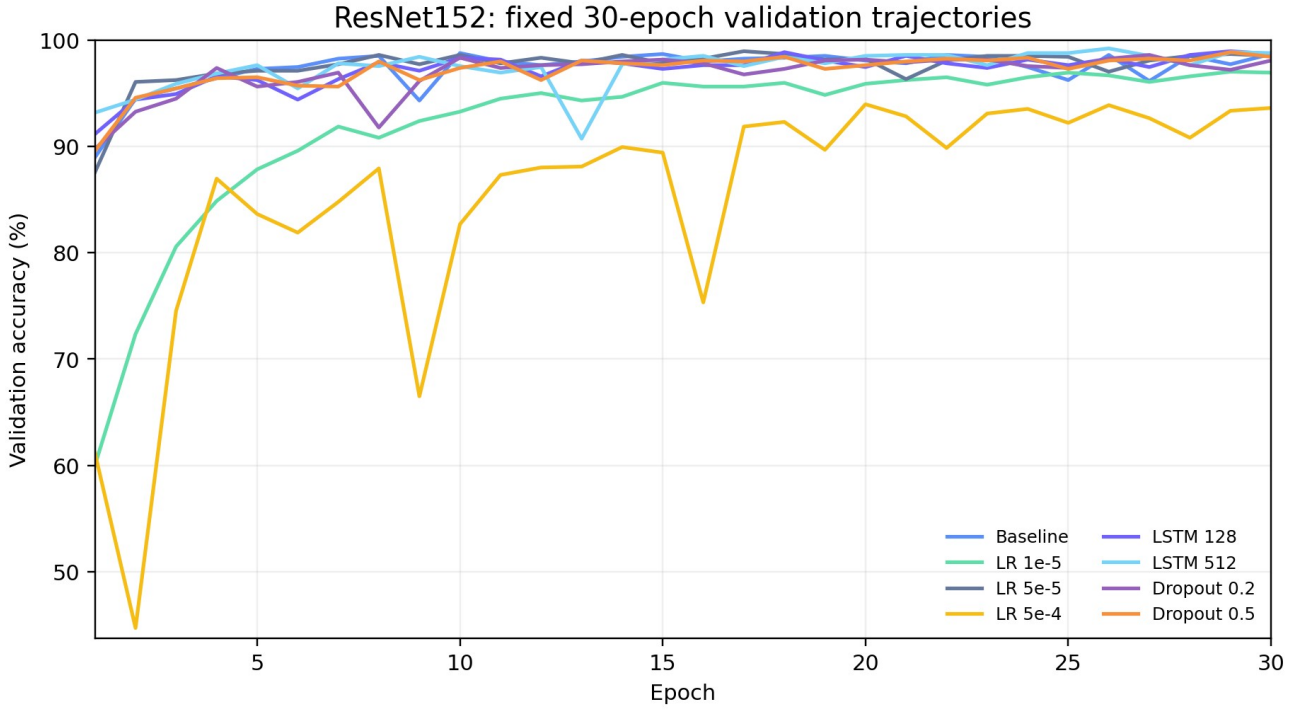


Figure B4. ResNet152 validation accuracy across the baseline and seven Phase 2B perturbations.

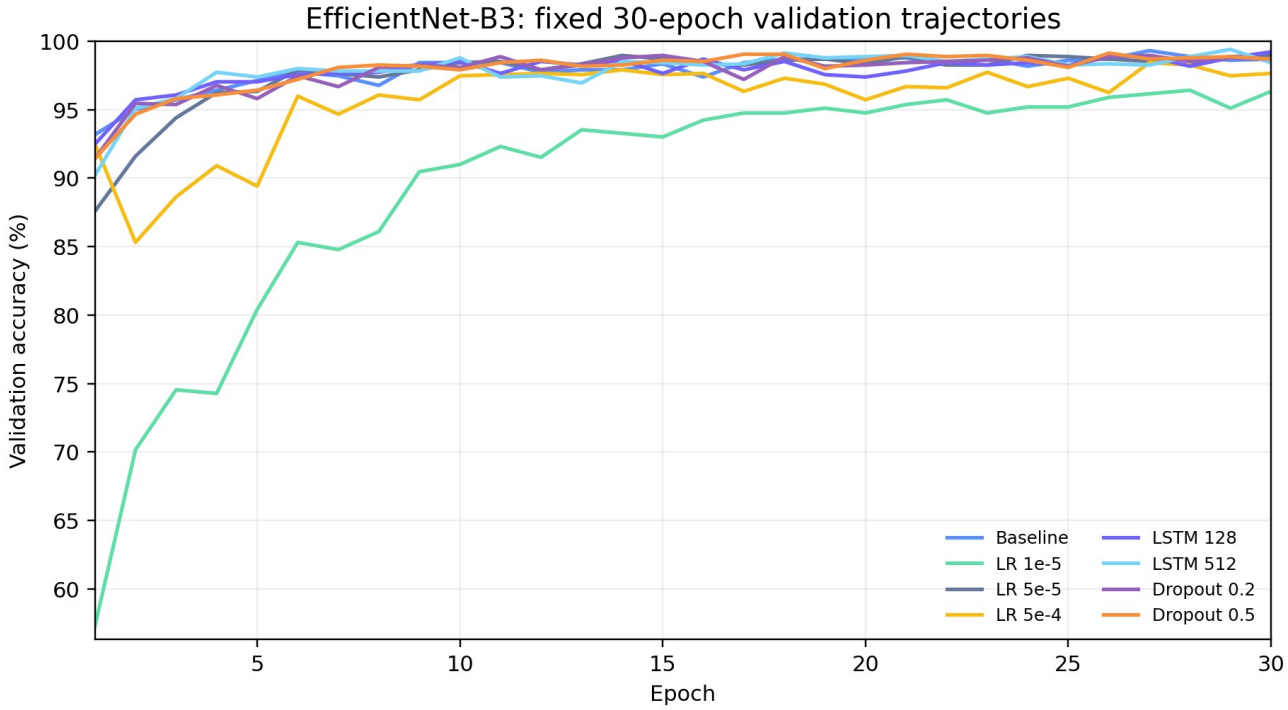


Figure B5. EfficientNet-B3 validation accuracy across the baseline and seven Phase 2B perturbations.

Appendix C. Evidence and artifact register

ID	Artifact/source	What it supports
E1	phase2a_baseline_results.zip	Five Phase 2A baseline training_results.json files; directly re-parsed for this report.
E2	phase2b_analysis_package.zip	35 Phase 2B training_results.json files + queue summary; directly re-parsed for this report.
E3	duplicate_leakage_audit.json	Authoritative audit counts/status and thresholds; project library source.
E4	Brain_Tumor_MRI_Classification_Final_Report.pdf	Phase 1 standardized test table, material_clean test table, explainability synthesis; project library source.
E5	03_PROJECT_JOURNEY_TIMELINE.md / 06_EXPERIMENT_HISTORY.md	Methodology evolution, Grad-CAM/occlusion protocols, leakage interpretation; project library sources.
E6	model_cnn_lstm.py / train_backbone.py / evaluate_backbone.py / GUI scripts	Architecture, training/evaluation behavior, deterministic feature extraction and GUI implementation.
E7	Current project-status handoff (2 Oct 2026)	Authority for completion status, Phase 2C priority, unresolved group-aware retraining, Review-I presentation convention and tool/workflow ownership.

Key project scripts / artifact families

- model_cnn_lstm.py - generic five-backbone CNN-LSTM architecture.
- train_backbone.py / train_backbone_hyperparams.py - baseline and sensitivity training.
- evaluate_neural_head.py - saved-checkpoint neural-head evaluation.
- summarize_neural_head_results.py - neural-head result aggregation.
- compare_neural_vs_pca_svm.py - downstream-classifier comparison.
- evaluate_backbone.py - deterministic deep-feature extraction, StandardScaler, PCA64, linear SVM evaluation.
- app.py + gui/* - Streamlit inference/performance/explainability GUI.
- gradcam_evaluation_app.py - frozen qualitative Grad-CAM protocol.
- occlusion_evaluation_app.py - final-SVM occlusion comparison.
- duplicate/leakage audit artifacts - exact/perceptual cross-split integrity check and material_clean subset definition.
- phase2/experiments_30epoch/* - Phase 2A fixed-30-epoch baselines.
- phase2/hyperparameter_dl/* - Phase 2B seven variants per backbone.

End-state summary

The project is no longer at baseline implementation. It is at the post-Phase-2B analysis stage. The next substantive scientific task is external validation, followed by an explicit decision on duplicate-group-aware re-splitting/retraining. Any future document should preserve the leakage caveat and distinguish historical contaminated-split performance, material_clean post-hoc evaluation, and Phase 2 validation sensitivity results.